Chapter 5: The Network Layer - Control Plane

1 5.1 Introduction

Control Plane Overview

Context: Forwarding/flow tables are the **principal elements** linking:

- **Data Plane** - forwards packets
- **Control Plane** - computes tables

Main Question: How are these tables computed, maintained & installed?
Two Key Tables:

- **Forwarding Table** → Destination-based forwarding
- **Flow Table** → Generalized forwarding (match-plus-action)

1.1 What These Tables Do

Forwarding Table

Purpose: Specifies local data-plane forwarding behavior
Function: Maps destination addresses → output ports
Used in: Traditional destination-based forwarding

Flow Table (Generalized Forwarding)

Match-Plus-Action Abstraction:

- **Match:** Identifies a flow based on header fields (L2, L3, L4).
- **Plus:** Additional internal data (ingress port, metadata).
- **Action:** What to do (Forward, Modify, Drop).

Actions supported:

- **Forward:** To specific output port(s).
- **Drop:** Discard the packet (e.g., Firewall).
- **Modify:** Rewrite headers (e.g., NAT, TTL).

- **Push/Pop:** Add/remove encapsulation (e.g., MPLS, VLAN).

Advantage: Consolidates middleboxes (Firewall, NAT, Load Balancer) into universal switches controlled by software.

Match-Plus-Action: Field Breakdown

What can we match on?

- **Link Layer (L2):** Source/Dest MAC, Ethernet Type, VLAN ID.
- **Network Layer (L3):** Source/Dest IP, IP Protocol, DSCP.
- **Transport Layer (L4):** Source/Dest Port (TCP/UDP).

This allows for "fine-grained" control beyond just destination IP.

2 Two Approaches to Network Control

2.1 1. Per-Router Control (Distributed)

Per-Router Control

Concept: Each router runs its own routing algorithm independently
Architecture: Distributed - routing & forwarding both inside each router

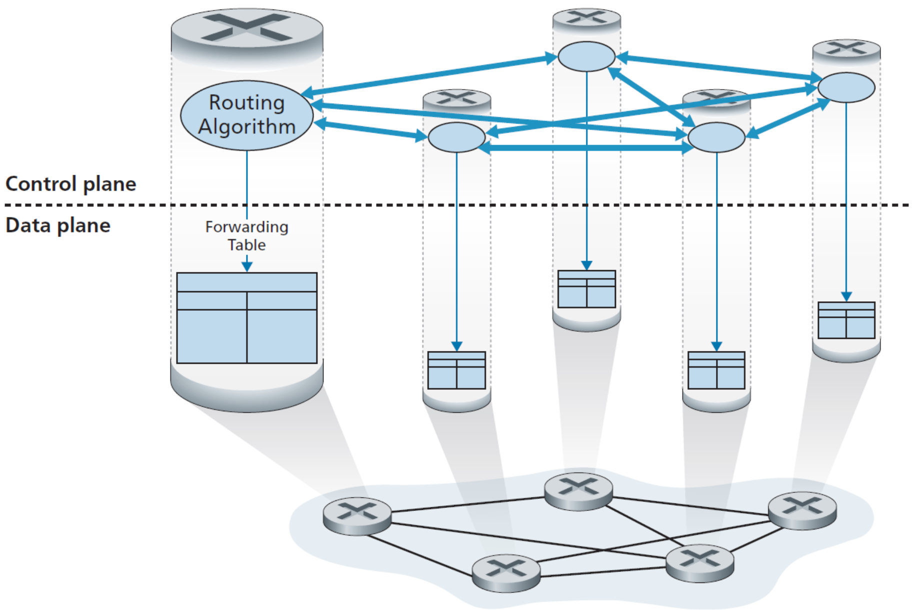

Figure 1: Per-router control: Individual routing algorithm components interact in the control plane

How It Works

Process:

1. Routing components in different routers **communicate with each other**
2. Each router **independently computes** its own forwarding table
3. Both control & data plane functions exist in **every router**

Key Point: Distributed intelligence - no central authority

Protocols Using This Approach

Used in Internet for decades:

- **OSPF (Open Shortest Path First)** - Section 5.3
  - Interior gateway protocol
  - Link-state routing algorithm
- **BGP (Border Gateway Protocol)** - Section 5.4
  - Exterior gateway protocol
  - Path-vector routing algorithm

2.2 2. Logically Centralized Control (SDN)

Logically Centralized Control

Concept: Central controller computes & distributes forwarding tables to all routers
Architecture: Centralized brain, distributed agents

How It Works

Components:

- **Controller (Brain):**
  - Computes & distributes forwarding/flow tables for ALL routers
  - Has global network view
  - Runs on remote server(s) (typically)
- **Control Agent (CA) in each router:**
  - Minimal functionality - just follows orders
  - Communicates with controller via **well-defined protocol**
  - Configures & manages local flow table

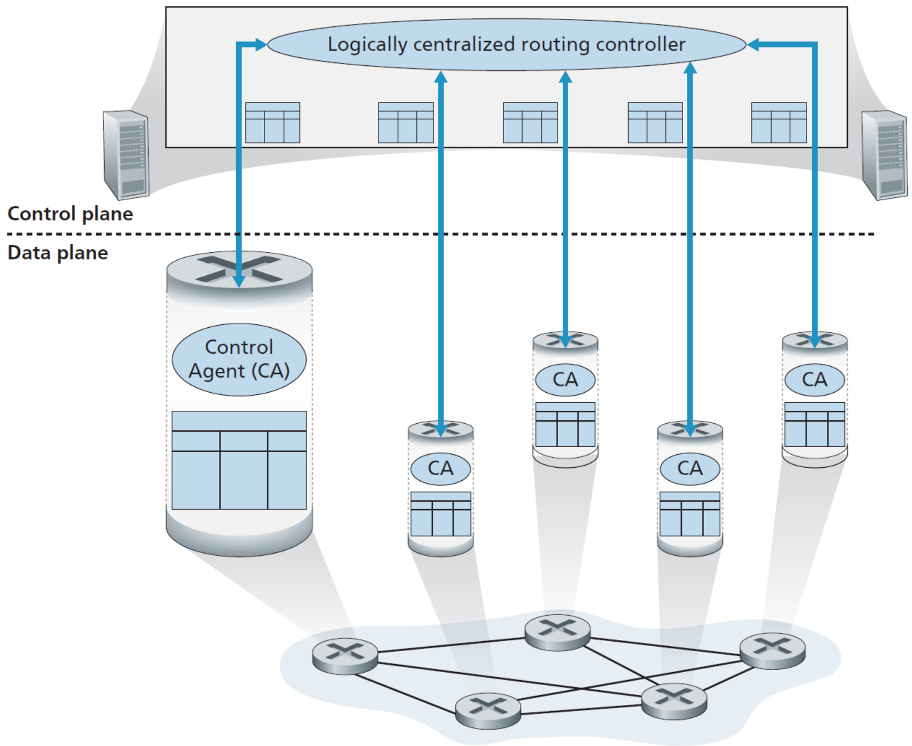

Figure 2: Logically centralized control: A distinct, typically remote, controller interacts with local control agents (CAs)

- Does NOT compute tables

Communication: Controller - CAs via protocol (CAs don't talk to each other)

Per-Router vs Centralized Control

Per-Router (Distributed)	**Logically Centralized (SDN)**
Intelligence: Distributed across every router.	**Intelligence:** Centralized in a remote controller.
Communication: Routers talk to neighbors (OSPF, BGP).	**Communication:** CAs talk only to controller (OpenFlow).
Computation: Each router runs routing algorithm.	**Computation:** Controller calculates for the whole path.
Data/Control: Bound together in the same hardware.	**Data/Control:** Explicitly decoupled/separated.
Innovation: Slow (requires industry-wide standards).	**Innovation:** Fast (simply update software on controller).

The Power of SDN: Traffic Engineering (TE)

Scenario: We want to route traffic from Source A to Dest B.

- **Traditional:** Routers use "Shortest Path First." If the shortest path is congested, traffic fails even if another path is idle.
- **SDN:** The controller sees the *entire network state*. It can route "Elephant Flows" (large data) over a longer but idle path, while keeping the shortest path for "Mice Flows" (latency-sensitive).

Benefit: Maximizes link utilization and avoids congestion automatically.

What "Logically Centralized" Means

Logically: Appears as single service point
Physically: Actually multiple servers for:

- Fault tolerance
- Scalability
- High availability

Think: One brain, multiple bodies

3 SDN in the Real World

Major SDN Deployments

Tech Giants:

- **Google B4** [Jain 2013]
  - SDN controls internal B4 global WAN
  - Interconnects data centers worldwide
- **Microsoft SWAN** [Hong 2013]
  - Manages routing between WAN - Data Center

ISPs:

- **COMCAST** - ActiveCore (actively integrating)
- **Deutsche Telecom** - Access 4.0 (actively integrating)
- **AT&T** [AT&T 2019] - *"75% of network functions fully virtualized & software-controlled"*

Telecom:

- **China Telecom & China Unicom** [Li 2015]

  - Within data centers (intra-datacenter)
  - Between data centers (inter-datacenter)
- **4G/5G Networks** - SDN is **central** (Chapter 8)

4 Generalized Forwarding & Match-Plus-Action

Match-Plus-Action Abstraction

What it enables:
Traditional:

- IP forwarding

Additional functions (previously needed separate middleboxes):

- **Load sharing/Load balancing**
- **Firewalling** - security filtering
- **NAT (Network Address Translation)**

Key advantage: Unified in SDN-enabled routers instead of separate devices

5 Key Takeaways

Summary

Two Control Approaches:

Per-Router (Traditional)	**Centralized (SDN)**
Distributed algorithms	Central controller
OSPF, BGP protocols	Software-defined
Each router computes	Controller computes all
Routers communicate	CAs don't communicate
Decades old	Modern approach

Industry Trend: Major shift to SDN happening NOW

- Google, Microsoft, AT&T already deployed
- ISPs actively integrating (COMCAST, Deutsche Telecom)
- Central to 4G/5G cellular networks
- China Telecom & Unicom using in data centers

Why SDN? Centralized control + programmability + global view = flexibility
Bottom line: SDN is not theory - it's production reality transforming networking